B13705023 蔡承叡 HW8

1. Server Setup

- `apt install -y slapd ldap-utils openssl` 下載 openLDAP server 和工具
- 在 "Configuring slapd" 的介面輸入 admin 密碼兩次
- `dpkg-reconfigure slapd` 重新 configure openldap，這個可以把前兩個設定都設好
    - Omit OpenLDAP Server? 選 no
    - DNS domain name 輸入 nasa.csie.ntu
    - Organization name 輸入 nasa
    - 密碼輸入兩次
    - Remove the database when slapd is purged? 預設(no, 都可以))
    - Move old database? Yes
- 現在可以 `systemctl start slapd`
- `nano add_orgunit.ldif`
- `ldapadd -x -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_orgunit.ldif`
- 驗證

- ```
  root@hw8-server:~# ldapsearch -x -H ldapi:/// -b dc=nasa,dc=csie,dc=ntu
  # extended LDIF
  #
  # LDAPv3
  # base <dc=nasa,dc=csie,dc=ntu> with scope subtree
  # filter: (objectclass=*)
  # requesting: ALL
  #

  # nasa.csie.ntu
  dn: dc=nasa,dc=csie,dc=ntu
  objectClass: top
  objectClass: dcObject
  objectClass: organization
  o: nasa
  dc: nasa

  # people, nasa.csie.ntu
  dn: ou=people,dc=nasa,dc=csie,dc=ntu
  objectClass: organizationalUnit
  ou: people

  # group, nasa.csie.ntu
  dn: ou=group,dc=nasa,dc=csie,dc=ntu
  objectClass: organizationalUnit
  ou: group

  # search result
  search: 2
  result: 0 Success

  # numResponses: 4
  # numEntries: 3
  ```

參考資料：
https://www.ibm.com/docs/zh-tw/power10?topic=commands-ldapsearch-command
https://documentation.ubuntu.com/server/how-to/openldap/install-openldap/

2. Client Setup

- `pacman -S openldap`
- `vim /etc/openldap/ldap.conf`
- 把 BASE 和 URI 改成 dc=nasa,dc=csie,dc=ntu 和 ldap://192.168.8.1
- 驗證
  ```

```
[root@hw8-client ~]# ldapsearch -x
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> (default) with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#

# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success

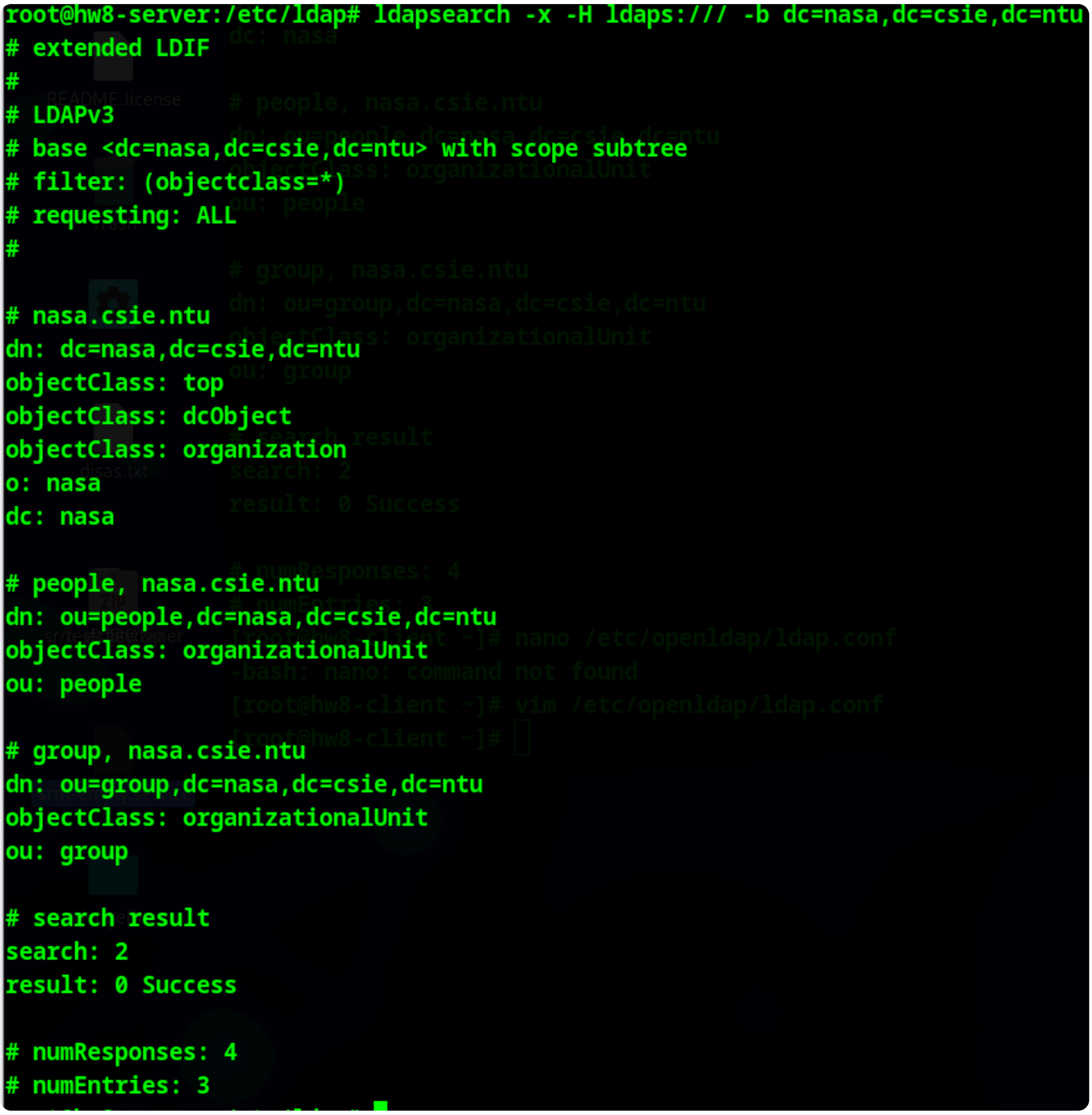
# numResponses: 4
# numEntries: 3
```

參考資料：
https://docs.google.com/presentation/d/1aJh_NVSIo39nRa7QAK99plr_uCQ9gJK_jf9P7wJK-1U/edit?slide=id.g32f400f0f0a_0_0#slide=id.g32f400f0f0a_0_0

3. LDAPS (LDAP over SSL)

(a)

On Server

- `cd /etc/ldap/sasl2`
- `openssl req -new -x509 -days 3650 -nodes -out ca-cert.pem -keyout ca-key.pem -subj "/C=TW/ST=Taipei/L=Taipei/O=NASA CSIE NTU/CN=NASA CA"` 生成 CA 的 private key 和憑證
- `openssl genrsa -out ldap-key.pem 2048` 生出 server private key
- `openssl req -new -key ldap-key.pem -out ldap-csr.pem -subj "/C=TW/ST=Taipei/L=Taipei/O=NASA CSIE NTU/CN=ldap.nasa.csie.ntu"` 生出 certificate signing request (CSR)
- `openssl x509 -req -days 3650 -in ldap-csr.pem -CA ca-cert.pem -CAkey ca-key.pem -CAcreateserial -out ldap-cert.pem` 用 CA 簽署伺服器憑證
- `chown openldap:openldap /etc/ldap/sasl2/*.pem`
- `chmod 600 /etc/ldap/sasl2/ldap-key.pem /etc/ldap/sasl2/ca-key.pem`
- `chmod 644 /etc/ldap/sasl2/ldap-cert.pem /etc/ldap/sasl2/ca-cert.pem`
- `nano tls_config.ldif` 在 /etc/ldap/ldifs 創建一個 tls config 的 ldif 檔，之後 ldifs 都會在這個資料夾
- `ldapmodify -Y EXTERNAL -H ldapi:/// -f tls_config.ldif` 套用 TLS config
- 去 /etc/default 然後 `nano slapd`
- 修改 `SLAPD_SERVICES="ldap:/// ldapi:/// ldaps:///"` (加入 ldaps:/// 啟用 LDAPS)
- `systemctl restart slapd`
  參考資料：
  https://learn.microsoft.com/zh-tw/troubleshoot/windows-server/active-directory/enable-ldap-over-ssl-3rd-certification-authority
  https://blog.miniasp.com/post/2019/02/25/Creating-Self-signed-Certificate-using-OpenSSL
  claude

(b)

- `nano ldap.conf` 在 /etc/ldap 修改 config 檔

- 加入 `TLS_CACERT /etc/ldap/sasl2/ca-cert.pem` , `TLS_REQCERT allow`

```
root@hw8-server:/etc/ldap# ldapsearch -x -ZZ -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#

# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 3
result: 0 Success

# numResponses: 4
# numEntries: 3
```

(c)

- `vim /etc/openldap/ldap.conf`
- 加入 `TLS_CACERT /etc/openldap/certs/ca-cert.pem` ,
  `TLS_REQCERT allow`

- 

```
root@hw8-server:/etc/ldap# ldapsearch -x -H ldaps:/// -b dc=nasa,dc=csie,dc=ntu
# extended LDIF
#
# LDAPv3
# base <dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#

# nasa.csie.ntu
dn: dc=nasa,dc=csie,dc=ntu
objectClass: top
objectClass: dcObject
objectClass: organization
o: nasa
dc: nasa

# people, nasa.csie.ntu
dn: ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: people

# group, nasa.csie.ntu
dn: ou=group,dc=nasa,dc=csie,dc=ntu
objectClass: organizationalUnit
ou: group

# search result
search: 2
result: 0 Success

# numResponses: 4
# numEntries: 3
```

(d)

- 直接在 /etc/default/slapd 把 `SLAPD_SERVICES="ldaps:/// ldapi:///"` 原本支援的 ldap:/// 去掉
- 到 client 端，在 /etc/openldap 多創一個 certs directory 然後進去
- `scp root@192.168.8.1:/etc/ldap/sasl2/ca-cert.pem .` 把 server 的 CA 憑證抓過來
- `systemctl restart slapd`

參考資料：
與蔡冠毅同學討論

(e)

- ```
  [root@hw8-client certs]# ldapsearch -x -H ldaps://192.168.8.1 -b dc=nasa,dc=csie
  ,dc=ntu
  # extended LDIF
  #
  # LDAPv3
  # base <dc=nasa,dc=csie,dc=ntu> with scope subtree
  # filter: (objectclass=*)
  # requesting: ALL
  #

  # nasa.csie.ntu
  dn: dc=nasa,dc=csie,dc=ntu
  objectClass: top
  objectClass: dcObject
  objectClass: organization
  o: nasa
  dc: nasa

  # people, nasa.csie.ntu
  dn: ou=people,dc=nasa,dc=csie,dc=ntu
  objectClass: organizationalUnit
  ou: people

  # group, nasa.csie.ntu
  dn: ou=group,dc=nasa,dc=csie,dc=ntu
  objectClass: organizationalUnit
  ou: group

  # search result
  search: 2
  result: 0 Success

  # numResponses: 4
  # numEntries: 3
  ```

(f)

- ```
  [root@hw8-client certs]# ldapsearch -x -H ldap://192.168.8.1 -b dc=nasa,dc=csie,
  dc=ntu
  ldap_sasl_bind(SIMPLE): Can't contact LDAP server (-1)
  ```

4. SSSD and Sudo

(a)

On Client

- `pacman -S sssd nss-pam-ldapd`
- `cd /etc/sssd`
- `vim sssd.conf` (content in ldif/conf file)
- `chmod 600 sssd.conf`
- `chown root:root sssd.conf`
- `cd /etc/pam.d`
- `cp system-auth system-auth.backup` 備份 pam.d 檔案
- `vim system-auth`
- 加入，success 每個 block 從高到低調整

```
auth        [success=1 default=ignore]  pam_sss.so  use_first_pass
account     [default=bad success=ok user_unknown=ignore] pam_sss.so
password    sufficient                  pam_sss.so          use_authtok
session     optional                    pam_sss.so
session     optional                    pam_mkhomedir.so     umask=0077
```

- 回到 /etc, `vim nsswitch.conf`
- 把原本 systemd 的地方全部替換成 sss
- `systemctl enable sssd`
- `systemctl start sssd`
- `cd ssh`
- `vim sshd_config`
- uncomment `PubkeyAuthentication yes` 和 `PasswordAuthentication yes` ，以及 UsePAM 改成 yes
- `systemctl restart sshd`

參考資料：
https://linuxeries.org/post/sssd/
https://linux.die.net/man/5/sssd.conf

(b)

> On Server

- `nano add_groups.ldif`

- `ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_groups.ldif` 匯入群組，注意這邊要多加 -H ldapi:/// (或是用 ldaps:///)
- ```
  root@hw8-server:~# ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu"
  -W -f add_groups.ldif
  Enter LDAP Password:
  adding new entry "cn=ta,ou=group,dc=nasa,dc=csie,dc=ntu"

  adding new entry "cn=student,ou=group,dc=nasa,dc=csie,dc=ntu"
  ```

On Client

- `cd /etc/sudoers.d`
- `vim ldap-users`
- 加入 %ta ALL=(ALL:ALL) ALL
- `chmod 440 ldap-users`

參考資料：
https://zhuanlan.zhihu.com/p/112200964

(c)

On Server

- `slappasswd` 兩次先幫 ta 和 student 生密碼hash，把 {SSHA}... 複製到 add_users.ldif 適合的地方
- `nano add_users.ldif` (gid 要跟剛剛創的 group 對應)
- `ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_users.ldif` 匯入成員到 LDAP (還沒到群組)
- ```
  root@hw8-server:~# ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu"
  -W -f add_users.ldif
  Enter LDAP Password:
  adding new entry "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu"

  adding new entry "uid=student01,ou=people,dc=nasa,dc=csie,dc=ntu"
  ```
- `nano add_members.ldif`
- `ldapmodify -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_members.ldif` 把成員加入 group

- ```
  root@hw8-server:~# ldapmodify -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_members.ldif
  Enter LDAP Password:
  modifying entry "cn=ta,ou=group,dc=nasa,dc=csie,dc=ntu"

  modifying entry "cn=student,ou=group,dc=nasa,dc=csie,dc=ntu"
  ```

參考資料：

https://www.thegeekstuff.com/2015/02/openldap-add-users-groups/

(d)

TA

- ```
  b13705023@nasaws1:~$ ssh -p 26844 ta01@localhost
  ta01@localhost's password:
  Creating directory '/home/ta01'.
  [ta01@hw8-client ~]$
  ```
- ```
  [ta01@hw8-client ~]$ sudo echo Hello World

  We trust you have received the usual lecture from the local System
  Administrator. It usually boils down to these three things:

      #1) Respect the privacy of others.
      #2) Think before you type.
      #3) With great power comes great responsibility.

  For security reasons, the password you type will not be visible.

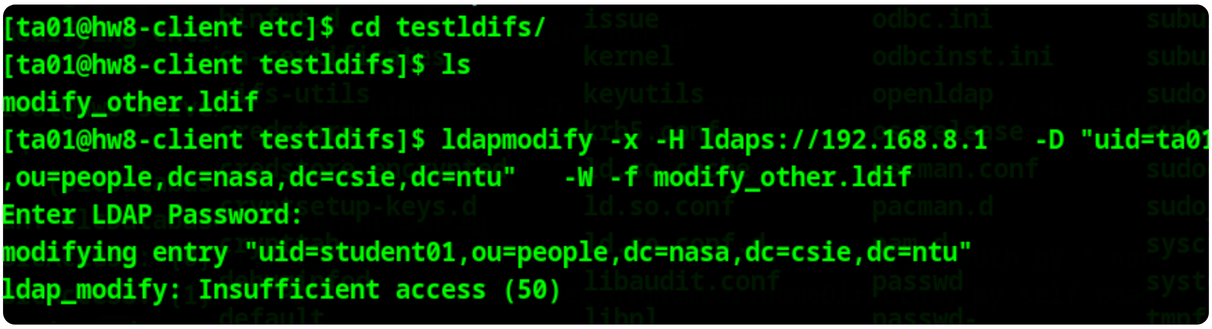
  [sudo] password for ta01:
  Hello World
  ```

Student

- ```
  b13705023@nasaws1:~$ ssh -p 26844 student01@localhost
  student01@localhost's password:
  Creating directory '/home/student01'.
  [student01@hw8-client ~]$ sudo echo Hello World
  ```

- ```
  [student01@hw8-client ~]$ sudo echo Hello World

  We trust you have received the usual lecture from the local System
  Administrator. It usually boils down to these three things:

      #1) Respect the privacy of others.
      #2) Think before you type.
      #3) With great power comes great responsibility.

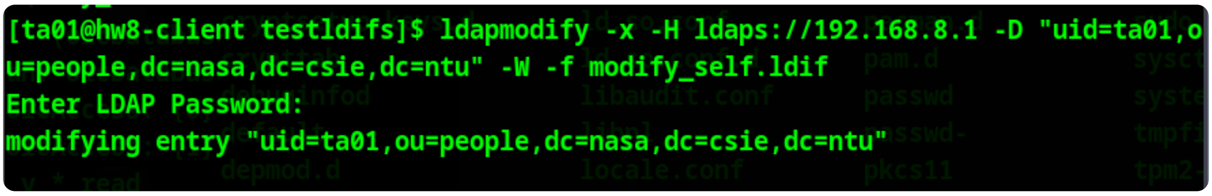
  For security reasons, the password you type will not be visible.

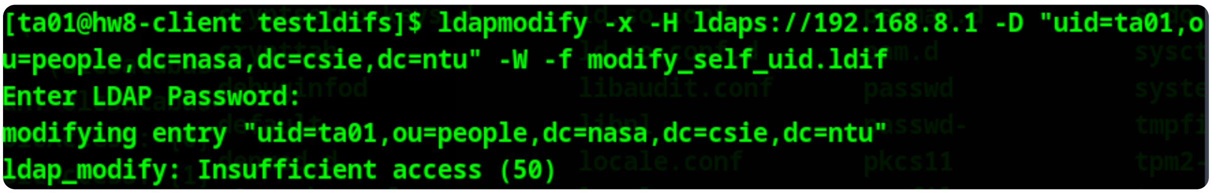
  [sudo] password for student01:
  student01 is not in the sudoers file.
  ```

5. ACL (Access Control Lists)

Setup

- `nano acl_conf.ldif`
- `ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f acl_conf.ldif modifying entry "olcDatabase={1}mdb,cn=config"` 套用新 acl config
- ```
  root@hw8-server:~# ldapmodify -Q -Y EXTERNAL -H ldapi:/// -f acl_conf.ldif
  modifying entry "olcDatabase={1}mdb,cn=config"
  ```
- ```
  root@hw8-server:~# sudo ldapsearch -Q -LLL -Y EXTERNAL -H ldapi:/// -b cn=config
   "(olcDatabase={1}mdb)" olcAccess
  dn: olcDatabase={1}mdb,cn=config
  olcAccess: {0}to attrs=userPassword by self write by anonymous auth by * none
  olcAccess: {1}to attrs=cn,uid,uidNumber,gidNumber,homeDirectory by self read b
   y * read
  olcAccess: {2}to * by self write by users read by anonymous read
  ```
- `systemctl restart slapd`

參考資料：
https://openldap.org/doc/admin24/access-control.html
claude

(a)

On Client

- `mkdir /etc/testldifs`
- `cd /etc/testldifs`
  ```

- `vim modify_other.ldif`
- `ldapmodify -x -H ldaps://192.168.8.1 -D "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu" -W -f modify_other.ldif`
- 

  顯示 ta 不能改 student 的屬性 (in this case: shell)

(b)

- `vim modify_self.ldif`
- `ldapmodify -x -H ldaps://192.168.8.1 -D "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu" -W -f modify_self.ldif`
- 

  顯示 ta 自己可以改自己改有權憲改的屬性
- `vim modify_self_uid.ldif`
- `ldapmodify -x -H ldaps://192.168.8.1 -D "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu" -W -f modify_self_uid.ldif`
- 

  顯示自己不能改自己的 uid

(c)

- `ldapsearch -x -H ldaps://192.168.8.1 -D "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu" -W -b "uid=student01,ou=people,dc=nasa,dc=csie,dc=ntu"`

- ```
  [ta01@hw8-client testldifs]$ ldapsearch -x -H ldaps://192.168.8.1 -D "uid=ta01,o
  u=people,dc=nasa,dc=csie,dc=ntu" -W -b "uid=student01,ou=people,dc=nasa,dc=csie,
  dc=ntu"
  Enter LDAP Password:
  # extended LDIF
  #
  # LDAPv3
  # base <uid=student01,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
  # filter: (objectclass=*)
  # requesting: ALL
  #

  # student01, people, nasa.csie.ntu
  dn: uid=student01,ou=people,dc=nasa,dc=csie,dc=ntu
  objectClass: inetOrgPerson
  objectClass: posixAccount
  objectClass: shadowAccount
  objectClass: top
  uid: student01
  cn: Student User One
  sn: User
  givenName: Student
  mail: student01@nasa.csie.ntu
  uidNumber: 10002
  gidNumber: 5002
  homeDirectory: /home/student01
  loginShell: /bin/bash
  description: Student User

  # search result
  search: 2
  result: 0 Success

  # numResponses: 2
  # numEntries: 1
  ```

  顯示用 ta 查找 student 是可行的，且不顯示密碼

- `ldapsearch -x -H ldaps://192.168.8.1 -b "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu"`
  ```

- ```
  [student01@hw8-client ~]$ ldapsearch -x -H ldaps://192.168.8.1 -b "uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu"
  # extended LDIF
  #
  # LDAPv3
  # base <uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
  # filter: (objectclass=*)
  # requesting: ALL
  #

  # ta01, people, nasa.csie.ntu
  dn: uid=ta01,ou=people,dc=nasa,dc=csie,dc=ntu
  objectClass: inetOrgPerson
  objectClass: posixAccount
  objectClass: shadowAccount
  objectClass: top
  uid: ta01
  cn: TA User One
  sn: User
  givenName: TA
  mail: ta01@nasa.csie.ntu
  uidNumber: 10001
  gidNumber: 5001
  homeDirectory: /home/ta01
  description: Teaching Assistant User
  loginShell: /bin/zsh

  # search result
  search: 2
  result: 0 Success

  # numResponses: 2
  # numEntries: 1
  ```
  顯示匿名查找也可，不會顯示密碼

6. LDAP Schema Extension

Backup

- `systemctl stop slapd`
- `cp -r /etc/ldap/slapd.d /etc/ldap/slapd.d.backup`
- `slapcat > /root/ldap_backup.ldif`
- `systemctl start slapd`

(a), (b), (c)

- `nano ntu_schema.ldif`
- `sudo ldapadd -Y EXTERNAL -H ldapi:/// -f ntu_schema.ldif`
  ```

- ```
  root@hw8-server:~# sudo ldapadd -Y EXTERNAL -H ldapi:/// -f ntu_schema.ldif
  SASL/EXTERNAL authentication started
  SASL username: gidNumber=0+uidNumber=0,cn=peercred,cn=external,cn=auth
  SASL SSF: 0
  adding new entry "cn=ntu,cn=schema,cn=config"
  ```

參考資料：

https://ldap.com/ldap-oid-reference-guide/

https://jumpcloud.com/it-index/what-are-ldap-object-classes

https://stackoverflow.com/questions/37908673/add-new-attribute-type-in-olc-schema-openldap

(d)

- `slappasswd` 生成給 ntu 學生密碼，一樣複製 {SSHA}..
- `nano add_ntu_std.ldif`
- `ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_ntu_std.ldif`
- ```
  root@hw8-server:~# sudo ldapadd -x -H ldapi:/// -D "cn=admin,dc=nasa,dc=csie,dc=ntu" -W -f add_ntu_std.ldif
  Enter LDAP Password:
  adding new entry "uid=ntu01,ou=people,dc=nasa,dc=csie,dc=ntu"
  ```
- (這邊原本一直遇到ldap_add: Invalid syntax (21)
  additional info: objectClass: value #3 invalid per syntax)，但把 schema 裡面的空格刪掉，然後重新匯入就成功拉

參考資料：

https://blog.csdn.net/weixin_41422086/article/details/106145226

與陳鼎元同學討論

(e)

- `ldapsearch -x -H ldaps://192.168.8.1 \ -b "uid=ntu01,ou=people,dc=nasa,dc=csie,dc=ntu"`

-

```
root@hw8-server:~# ldapsearch -x -H ldaps://192.168.8.1 \
  -b "uid=ntu01,ou=people,dc=nasa,dc=csie,dc=ntu"
# extended LDIF
#
# LDAPv3
# base <uid=ntu01,ou=people,dc=nasa,dc=csie,dc=ntu> with scope subtree
# filter: (objectclass=*)
# requesting: ALL
#

# ntu01, people, nasa.csie.ntu
dn: uid=ntu01,ou=people,dc=nasa,dc=csie,dc=ntu
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
objectClass: ntuStudent
uid: ntu01
cn: NTU Student One
sn: Student
givenName: NTU
mail: ntu01@ntu.edu.tw
uidNumber: 10003
gidNumber: 5002
homeDirectory: /home/ntu01
loginShell: /bin/bash
description: NTU Student with custom schema
studentEntryMethod: GSAT/INTERVIEW
studentAdvisor: Pf. Michael Tsai
studentClub: AI club
studentClub: Piano club

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1
```